
1.

The pH at equivalence point in strong acid–strong base titration is:

- A) < 7
- B) > 7
- C) $= 7$
- D) Depends on K_a

👉 Answer: C

2.

Which indicator is suitable for weak acid–strong base titration?

- A) Methyl orange
- B) Phenolphthalein
- C) Both
- D) None

👉 Answer: B

3.

At half equivalence point:

- A) $\text{pH} = \text{pK}_b$
- B) $\text{pH} = \text{pK}_a$
- C) $\text{pH} = 7$
- D) $\text{pH} = 0$

👉 Answer: B

4.

Equivalent weight of H_2SO_4 is:

- A) MW
- B) $\text{MW}/2$

- C) $MW/3$
- D) $2 \times MW$

👉 Answer: B

5.

A buffer solution resists changes in:

- A) Temperature
- B) Volume
- C) pH
- D) Pressure

👉 Answer: C

6.

Which solution is acidic?

- A) NaCl
- B) NH_4Cl
- C) CH_3COONa
- D) NaOH

👉 Answer: B

7.

K_w at $25^\circ C$ equals:

- A) 10^{-7}
- B) 10^{-14}
- C) 10^7
- D) 1

👉 Answer: B

8.

Which has buffer region?

- A) Strong acid + strong base
- B) Weak acid + strong base
- C) Weak + weak
- D) None

👉 Answer: B

9.

Primary standard must be:

- A) Impure
- B) Unstable
- C) Highly pure
- D) Volatile

👉 Answer: C

10.

pH of acidic solution is:

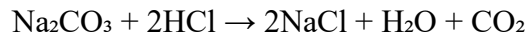
- A) > 7
- B) $= 7$
- C) < 7
- D) $= 14$

👉 Answer: C

Problem 1

20 ml of Na_2CO_3 solution required 15 ml of 0.1 N HCl. Calculate concentration (g/L).

✓ Step 1: Write equation



✓ Step 2: Calculate MW

$$\text{Na}_2\text{CO}_3 = 106$$

✓ Step 3: Equivalent weight

$$\text{Eq. wt} = 106 / 2 = 53$$

✓ Step 4: Equivalent factor (F)

$$0.1 \text{ eq} = 53 / 10 = 5.3 \text{ g/L}$$

$$F = 5.3 / 1000 = 0.0053 \text{ g}$$

✓ Step 5: Calculate amount

$$\begin{aligned} \text{Amount} &= V \times F \\ &= 15 \times 0.0053 = 0.0795 \text{ g} \end{aligned}$$

✓ Step 6: Concentration (g/ml)

$$= 0.0795 / 20 = 0.003975 \text{ g/ml}$$

✓ Step 7: Convert to g/L

$$= 0.003975 \times 1000$$

$$\text{👉} = 3.975 \text{ g/L}$$

Problem 2

25 ml sample consumed 10 ml of 0.1 N HCl. Find concentration.

Step 1:

$$F = 0.0053$$

Step 2:

$$\text{Amount} = 10 \times 0.0053 = 0.053 \text{ g}$$

Step 3:

$$\text{Concentration} = 0.053 / 25 = 0.00212 \text{ g/ml}$$

Step 4:

$$= 0.00212 \times 1000$$

$$\text{👉} = 2.12 \text{ g/L}$$

Problem 3 (With factor 🙌)

20 ml sample consumed 12 ml HCl, f = 1.02

Step 1:

$$F = 0.0053$$

Step 2:

$$\text{Amount} = 12 \times 1.02 \times 0.0053$$

$$= 0.0649 \text{ g}$$

Step 3:

$$\text{Concentration} = 0.0649 / 20$$

$$= 0.003245 \text{ g/ml}$$

Step 4:

$$= 0.003245 \times 1000$$

$$\text{👉} = 3.245 \text{ g/L}$$

Problem 4 (Purity 🧤)

0.5 g sample required 10 ml 0.1 N HCl. Calculate % purity.

Step 1:

$$F = 0.0053$$

Step 2:

$$\text{Amount} = 10 \times 0.0053 = 0.053 \text{ g}$$

Step 3:

% purity:

$$= (0.053 \times 100) / 0.5$$

$$= 10.6 \%$$

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1.

At equivalence point of weak acid–strong base titration:

- A) $[H^+] = [OH^-]$
- B) $pH = 7$
- C) $pH > 7$
- D) $pH < 7$

 **Answer: C**

2.

Which of the following has the highest buffer capacity?

- A) 0.1 M acid + 0.01 M salt
- B) 0.1 M acid + 0.1 M salt
- C) 0.01 M acid + 0.01 M salt
- D) 0.2 M acid only

 **Answer: B**

3.

If $pH = pK_a$, then:

- A) $[Acid] = 0$
- B) $[Salt] = 0$
- C) $[Salt] = [Acid]$
- D) $pH = 7$

 **Answer: C**

4.

Which salt gives basic solution?

- A) NH_4Cl
- B) NaCl
- C) CH_3COONa
- D) HCl

👉 Answer: C

5.

In strong acid–weak base titration, the suitable indicator is:

- A) Phenolphthalein
- B) Methyl orange
- C) Both
- D) None

👉 Answer: B

6.

If K_a increases, the acid becomes:

- A) weaker
- B) stronger
- C) neutral
- D) basic

👉 Answer: B

7.

Which statement is correct?

- A) Strong acids partially dissociate
- B) Weak acids fully dissociate
- C) Strong acids fully dissociate
- D) Weak bases fully dissociate

👉 Answer: C

8.

Which curve shows no sharp equivalence point?

- A) Strong + Strong
- B) Weak + Strong
- C) Strong + Weak
- D) Weak + Weak

👉 Answer: D

9.

Buffer solution contains:

- A) Strong acid + strong base
- B) Weak acid + its salt
- C) Strong acid only
- D) Strong base only

👉 Answer: B

10.

pH + pOH equals:

- A) 7
- B) 1
- C) 10
- D) 14

👉 Answer: D

Problem 5

30 ml of Na_2CO_3 consumed 20 ml of 0.1 N HCl. Calculate concentration (g/L).

Step 1:

$$F = 0.0053$$

Step 2:

$$\text{Amount} = 20 \times 0.0053 = 0.106 \text{ g}$$

Step 3:

$$\text{Concentration (g/ml)} = 0.106 / 30 = 0.00353$$

Step 4:

$$\text{g/L} = 0.00353 \times 1000$$

$$\text{👉} = 3.53 \text{ g/L}$$

Problem 6 (Reverse)

Calculate volume of 0.1 N HCl needed to neutralize 25 ml Na_2CO_3 (3.18 g/L).

Step 1:

$$\begin{aligned} &\text{Convert concentration} \rightarrow \text{g/ml} \\ &= 3.18 / 1000 = 0.00318 \text{ g/ml} \end{aligned}$$

Step 2:

$$\text{Amount in 25 ml:} = 25 \times 0.00318 = 0.0795 \text{ g}$$

Step 3:

$$F = 0.0053$$

Step 4:

$$\begin{aligned} \text{Volume} &= \text{Amount} / F \\ &= 0.0795 / 0.0053 \end{aligned}$$

$$\text{👉} = 15 \text{ ml}$$



Problem 7 (Factor + tricky 🤔)

20 ml sample consumed 10 ml HCl, $f = 0.98$

Step 1:

$$F = 0.0053$$

Step 2:

$$\begin{aligned} \text{Amount} &= 10 \times 0.98 \times 0.0053 \\ &= 0.05194 \text{ g} \end{aligned}$$

Step 3:

$$\begin{aligned}\text{Concentration} &= 0.05194 / 20 \\ &= 0.002597 \text{ g/ml}\end{aligned}$$

Step 4:

$$\text{g/L} = 2.597 \text{ g/L}$$

Problem 8 (Purity)

1 g impure Na_2CO_3 required 20 ml of 0.1 N HCl. Find % purity.

Step 1:

$$F = 0.0053$$

Step 2:

$$\text{Amount} = 20 \times 0.0053 = 0.106 \text{ g}$$

Step 3:

% purity:

$$\begin{aligned}&= (0.106 \times 100) / 1 \\ &= 10.6 \%\end{aligned}$$

Problem 9 (Different volume)

50 ml sample required 25 ml HCl

Step 1:

$$\text{Amount} = 25 \times 0.0053 = 0.1325 \text{ g}$$

Step 2:

$$\text{Concentration} = 0.1325 / 50 = 0.00265 \text{ g/ml}$$

Step 3:

$$\text{g/L} = 2.65 \text{ g/L}$$